

Lecture 9

ER- and EER-to-Relational Mapping

- Reference: Chapter 9

- **Step 1: Mapping of Regular Entity Types.**
 - For each regular (strong) entity type E in the ER schema, create a relation R that includes all the simple attributes of E.
 - Choose one of the key attributes of E as the primary key for R.
 - If the chosen key of E is composite, the set of simple attributes that form it will together form the primary key of R.

Example: We create the relations EMPLOYEE, DEPARTMENT, and PROJECT in the relational schema corresponding to the regular entities in the ER diagram.

- SSN, DNUMBER, and PNUMBER are the primary keys for the relations EMPLOYEE, DEPARTMENT, and PROJECT as shown.

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary
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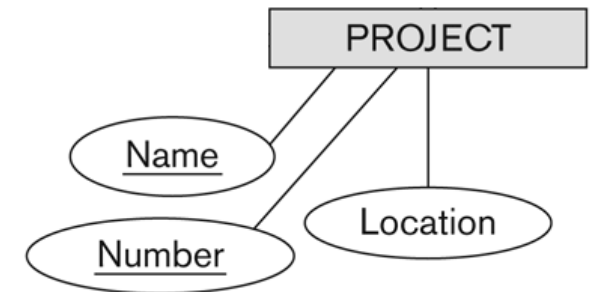
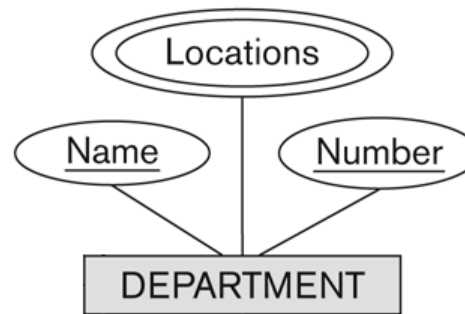
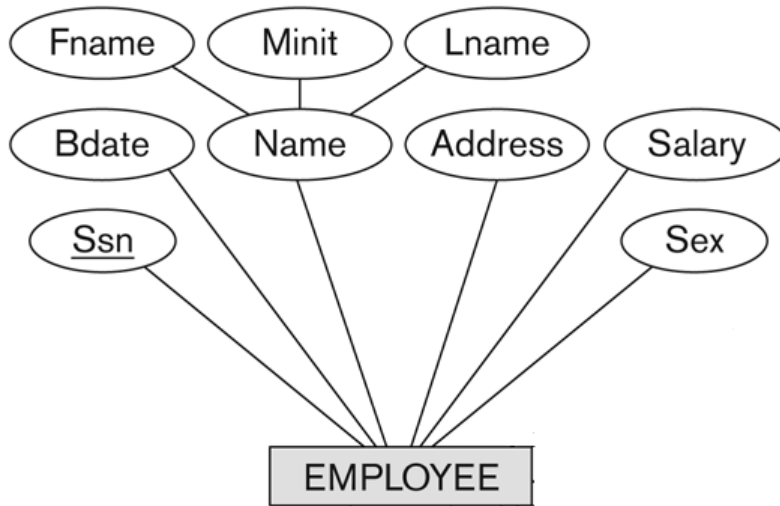
DEPARTMENT

Dname	<u>Dnumber</u>
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PROJECT

Pname	<u>Pnumber</u>	Plocation
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ER-to-Relational Mapping Algorithm



EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary
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DEPARTMENT

Dname	<u>Dnumber</u>
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PROJECT

Pname	<u>Pnumber</u>	Plocation
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• Step 2: Mapping of Weak Entity Types

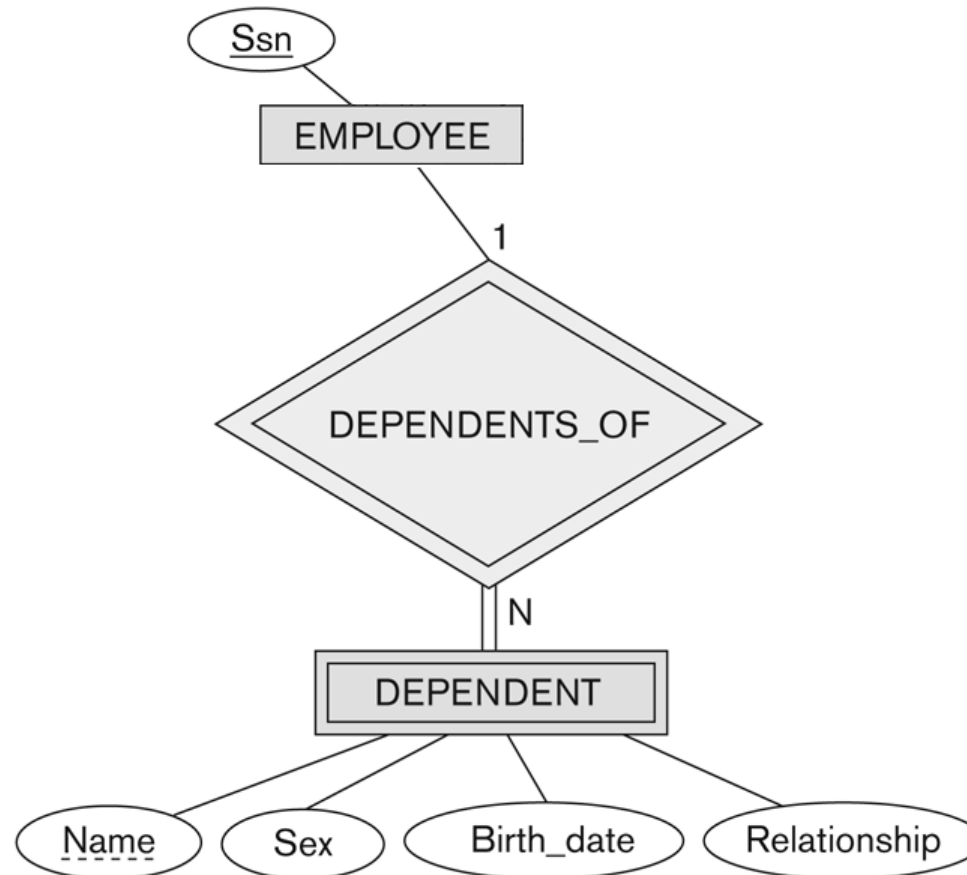
- For each weak entity type W in the ER schema with owner entity type E , create a relation R & include all simple attributes (or simple components of composite attributes) of W as attributes of R .
- Also, include as foreign key attributes of R the primary key attribute(s) of the relation(s) that correspond to the owner entity type(s).
- The primary key of R is the *combination* of the primary key(s) of the owner(s) and the partial key of the weak entity type W , if any.

Example: Create the relation DEPENDENT in this step to correspond to the weak entity type DEPENDENT.

- Include the primary key SSN of the EMPLOYEE relation as a foreign key attribute of DEPENDENT (renamed to ESSN).
- The primary key of the DEPENDENT relation is the combination {ESSN, DEPENDENT_NAME}.

DEPENDENT

<u>Essn</u>	<u>Dependent_name</u>	Sex	Bdate	Relationship
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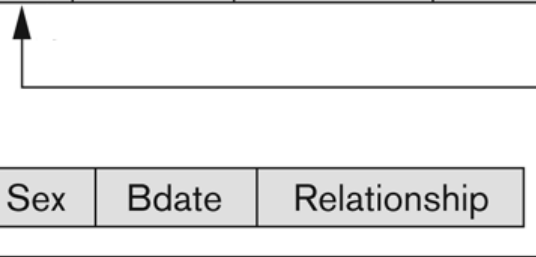


EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary
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DEPENDENT

<u>Essn</u>	<u>Dependent_name</u>	Sex	Bdate	Relationship
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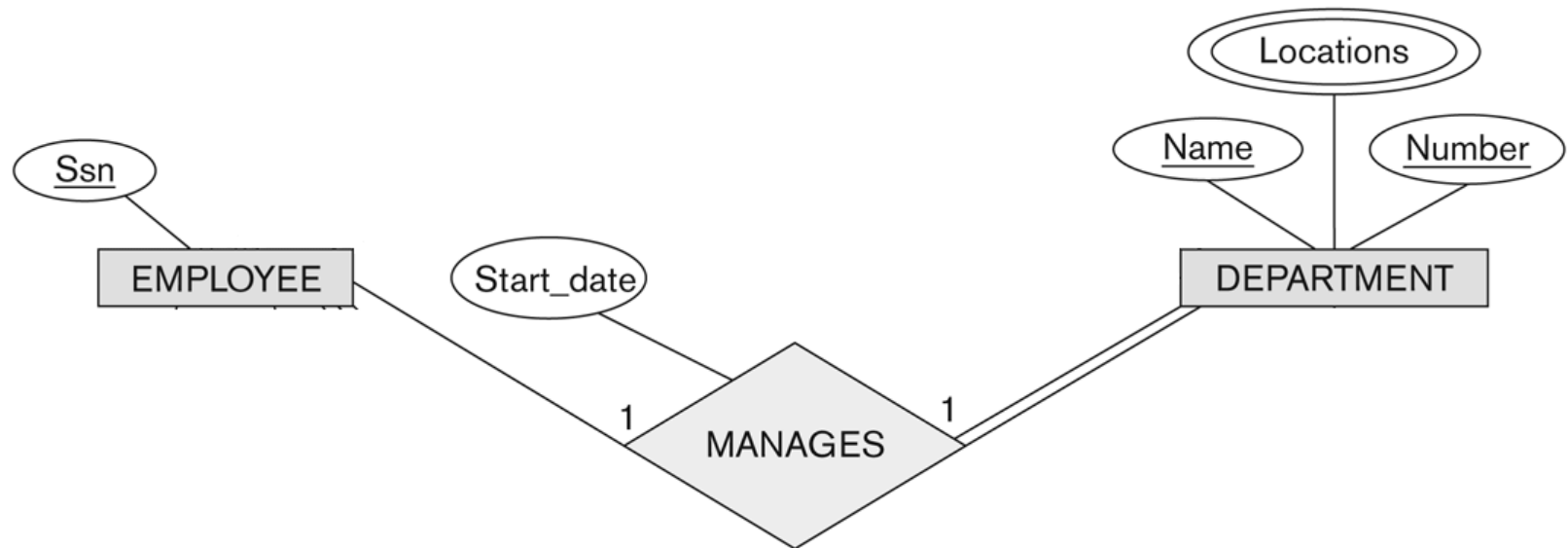


- **Step 3: Mapping of Binary 1:1 Relation Types**
 - For each binary 1:1 relationship type R in the ER schema, identify the relations S and T that correspond to the entity types participating in R.

ER-to-Relational Mapping Algorithm (3)

- There are three possible approaches:
 1. **Foreign Key approach:** Choose one of the relations-say S-and include a foreign key in S the primary key of T. It is better to choose an entity type with total participation in R in the role of S.
 - Example: 1:1 relation MANAGES is mapped by choosing the participating entity type DEPARTMENT to serve in the role of S, because its participation in the MANAGES relationship type is total.
 2. **Merged relation option:** An alternate mapping of a 1:1 relationship type is possible by merging the two entity types and the relationship into a single relation. This may be appropriate when both participations are total.
 3. **Cross-reference or relationship relation option:** The third alternative is to set up a third relation R for the purpose of cross-referencing the primary keys of the two relations S and T representing the entity types.

ER-to-Relational Mapping Algorithm (3)



EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary
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DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
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- **Step 4: Mapping of Binary 1:N Relationship Types.**

- For each regular binary 1:N relationship type R, identify the relation S that represent the participating entity type at the N-side of the relationship type.
- Include as foreign key in S the primary key of the relation T that represents the other entity type participating in R.
- Include any simple attributes of the 1:N relation type as attributes of S.

ER-to-Relational Mapping Algorithm (4)

- Example: 1:N relationship types SUPERVISION, WORKS_FOR and CONTROLS in the figure.
 - For WORKS_FOR we include the primary key DNUMBER of the DEPARTMENT relation as foreign key in the EMPLOYEE relation and call it DNO.

EMPLOYEE

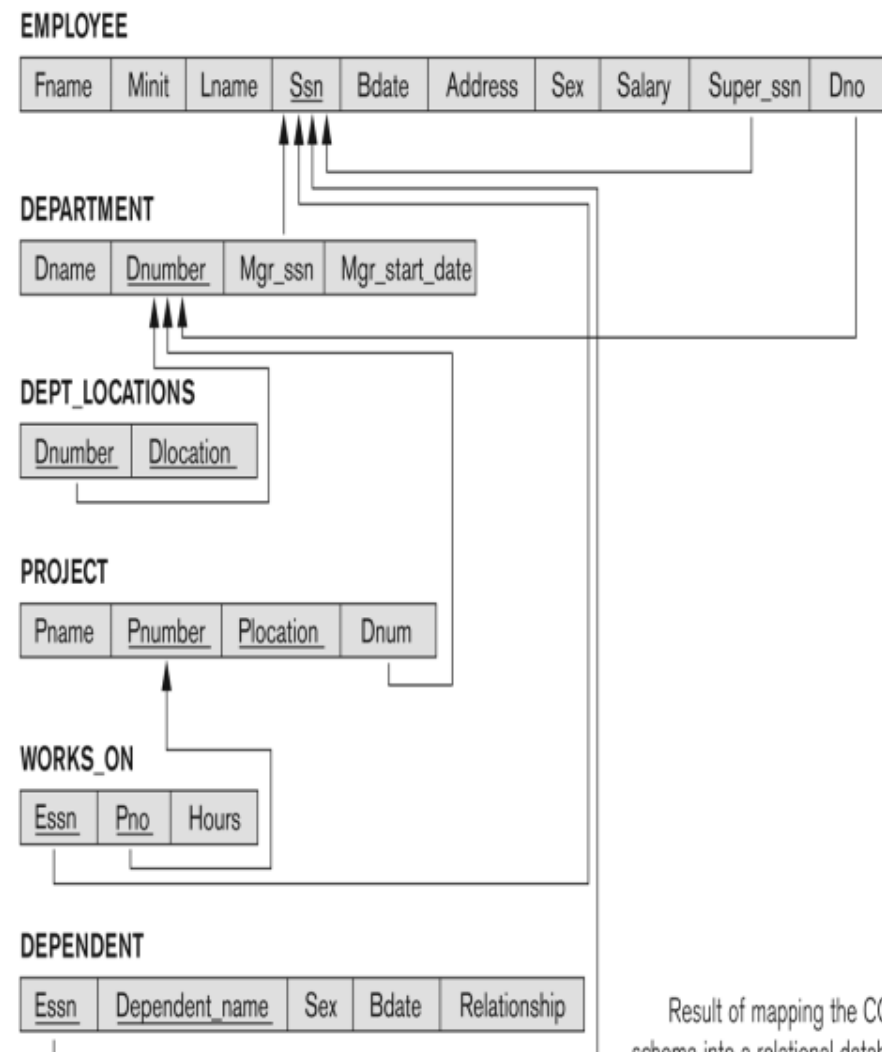
Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn
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EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
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PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
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- **Step 5: Mapping of Binary M:N Relationship Types.**

- For each regular binary M:N relationship type R, *create a new relation S* to represent R.
- Include as foreign key attributes in S the primary keys of the relations that represent the participating entity types; *their combination will form the primary key* of S.
- Also include any simple attributes of the M:N relationship type (or simple components of composite attributes) as attributes of S.

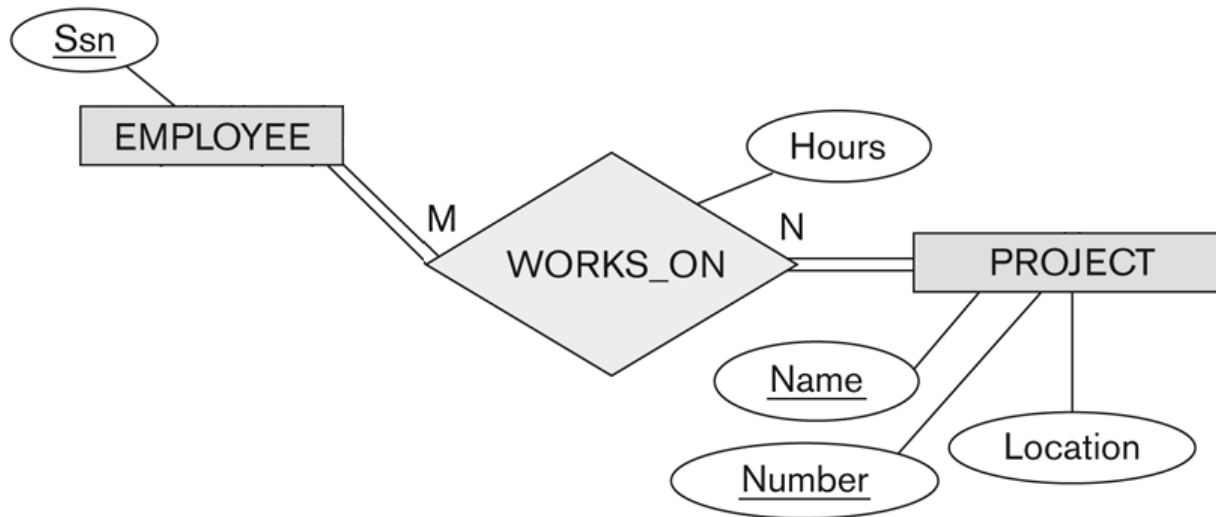
ER-to-Relational Mapping Algorithm (5)

- Example: The M:N relationship type WORKS_ON from the ER diagram is mapped by creating a relation WORKS_ON in the relational database schema.
 - The primary keys of the PROJECT and EMPLOYEE relations are included as foreign keys in WORKS_ON and renamed PNO and ESSN, respectively.
 - The primary key of the WORKS_ON relation is the combination of the foreign key attributes {ESSN, PNO}.

WORKS_ON

<u>Essn</u>	<u>Pno</u>	Hours
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ER-to-Relational Mapping Algorithm (5)



EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
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PROJECT

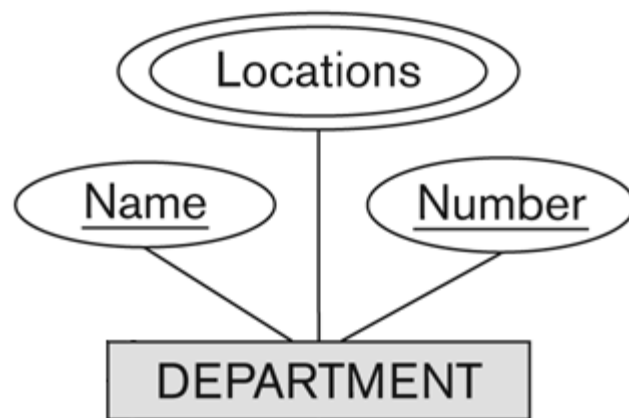
Pname	<u>Pnumber</u>	Plocation	Dnum
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WORKS_ON

<u>Essn</u>	<u>Pno</u>	Hours
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- **Step 6: Mapping of Multivalued attributes.**
 - For each multivalued attribute A, create a new relation R.
 - This relation R will include an attribute corresponding to A, plus the primary key attribute K-as a foreign key in R-of the relation that represents the entity type of relationship type that has A as an attribute.
 - The primary key of R is the combination of A and K. If the multivalued attribute is composite, we include its simple components.

- **Example:** The relation DEPT_LOCATIONS is created.
 - The attribute DLOCATION represents the multivalued attribute LOCATIONS of DEPARTMENT, while DNUMBER-as foreign key-represents the primary key of the DEPARTMENT relation.
 - The primary key of R is the combination of {DNUMBER, DLOCATION}.



DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
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ER-to-Relational Mapping Algorithm (7)

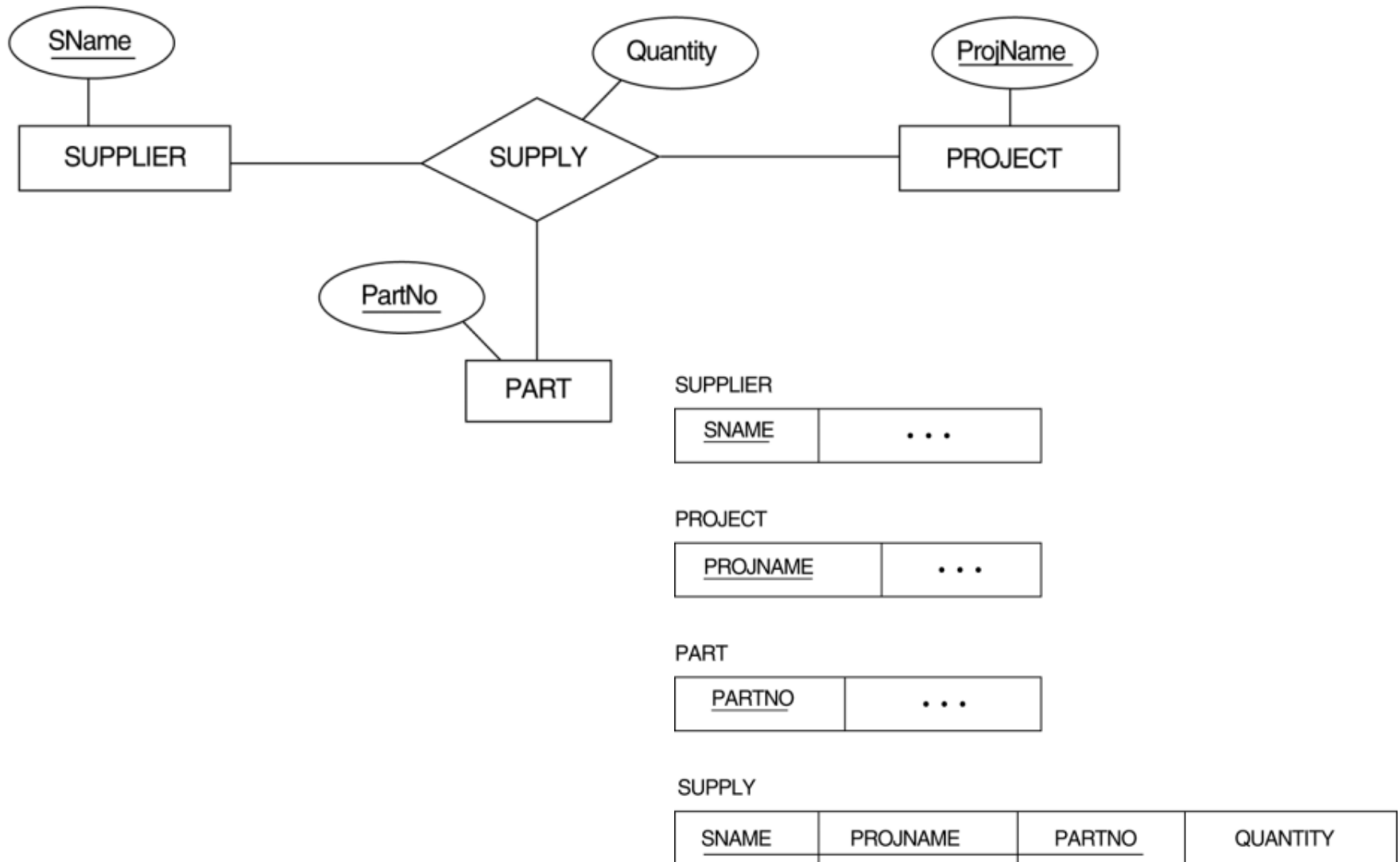
- **Step 7: Mapping of N-ary Relationship Types.**

- For each n-ary relationship type R, where $n > 2$, create a new relationship S to represent R.
- Include as foreign key attributes in S the primary keys of the relations that represent the participating entity types.
- Also include any simple attributes of the n-ary relationship type (or simple components of composite attributes) as attributes of S.

- **Example:** The relationship type SUPPY in the ER on the next slide.

- This can be mapped to the relation SUPPLY shown in the relational schema, whose primary key is the combination of the three foreign keys {SNAME, PARTNO, PROJNAME}

Mapping the *n*-ary relationship type SUPPLY

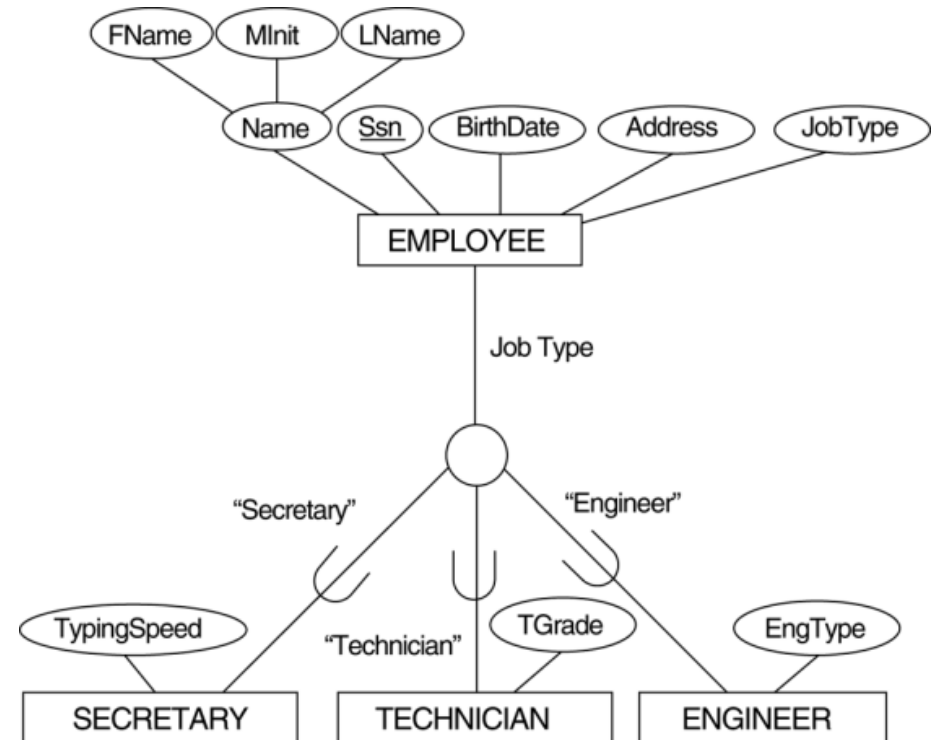


Mapping EER Model Constructs to Relations (8)

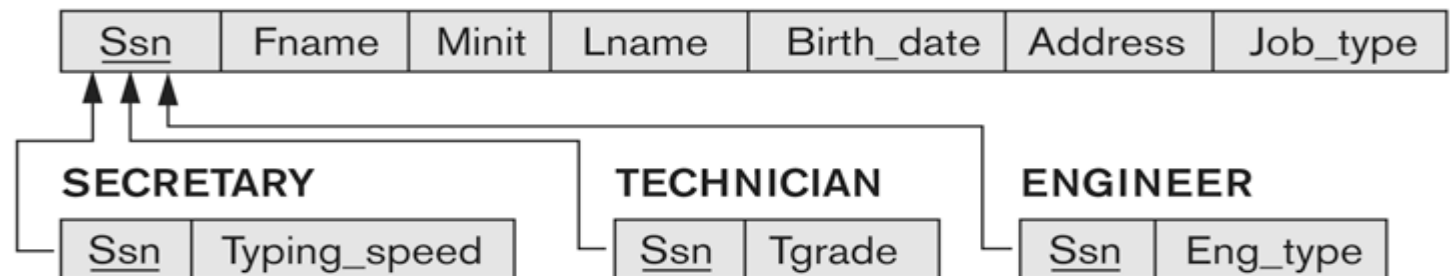
- **Step8: Options for Mapping Specialization or Generalization.**
 - Convert each specialization with m subclasses $\{S_1, S_2, \dots, S_m\}$ and generalized superclass C , where the attributes of C are $\{k, a_1, \dots, a_n\}$ and k is the (primary) key, into relational schemas using one of the four following options:
 - Option 8A: Multiple relations-Superclass and subclasses
 - Option 8B: Multiple relations-Subclass relations only
 - Option 8C: Single relation with one type attribute
 - Option 8D: Single relation with multiple type attributes

Mapping EER Model Constructs to Relations (8)

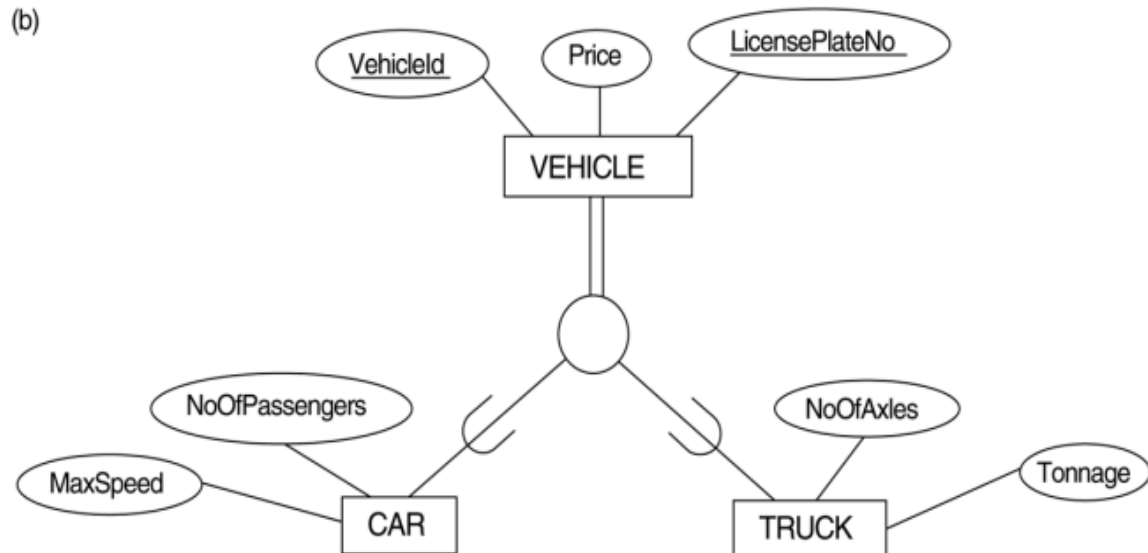
- Option 8A: Multiple relations-
Superclass and subclasses



(a) EMPLOYEE



- Option 8B: Multiple relations-Subclass relations only (b)



(b) CAR

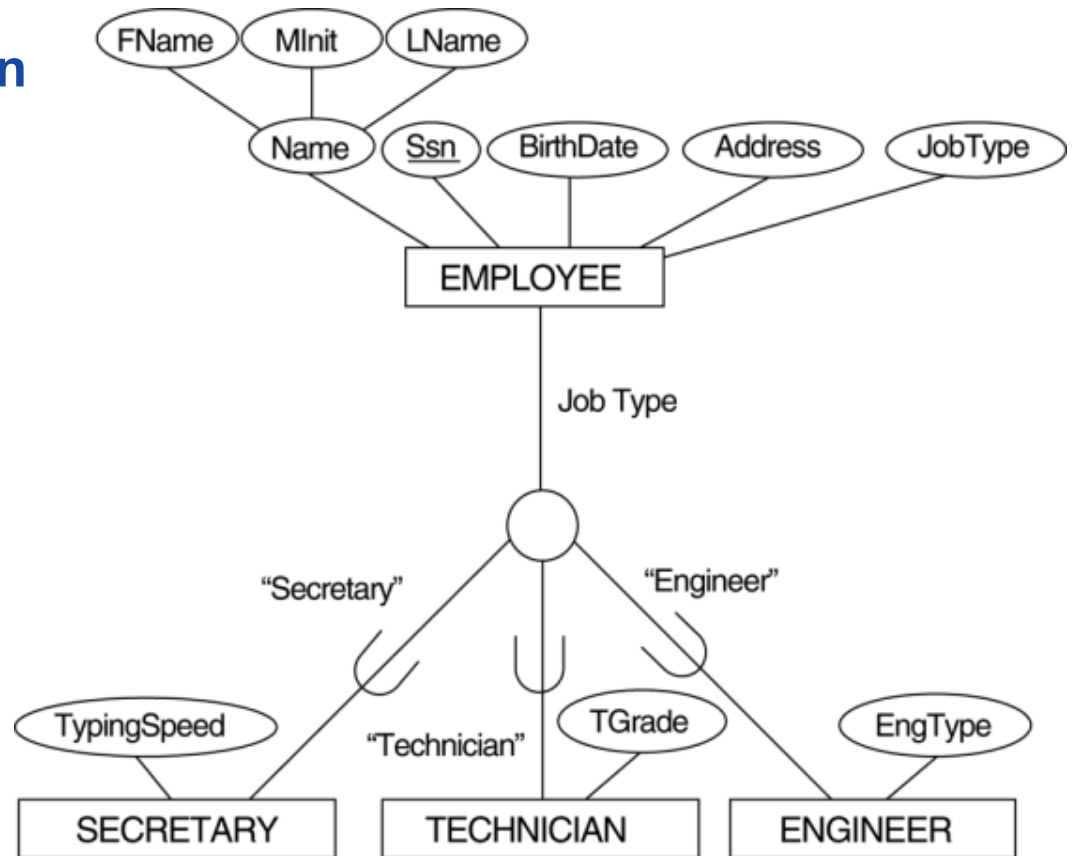
<u>VehicleId</u>	LicensePlateNo	Price	MaxSpeed	NoOfPassengers
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TRUCK

<u>VehicleId</u>	LicensePlateNo	Price	NoOfAxes	
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Mapping EER Model Constructs to Relations (8)

- Option 8C: Single relation with one type attribute

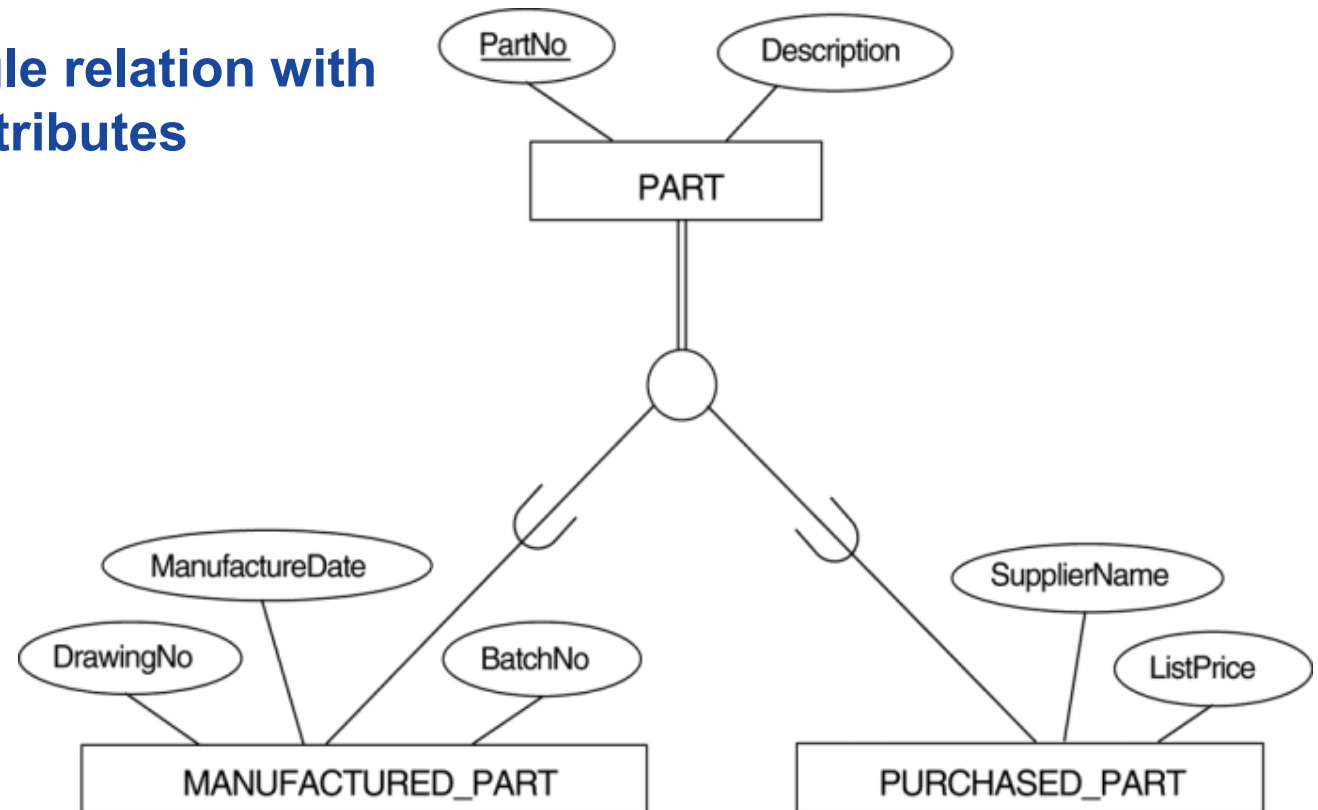


(c) EMPLOYEE

<u>SSN</u>	FName	MInit	LName	BirthDate	Address	JobType	TypingSpeed	TGrade	
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Mapping EER Model Constructs to Relations (8)

- Option 8D: Single relation with multiple type attributes

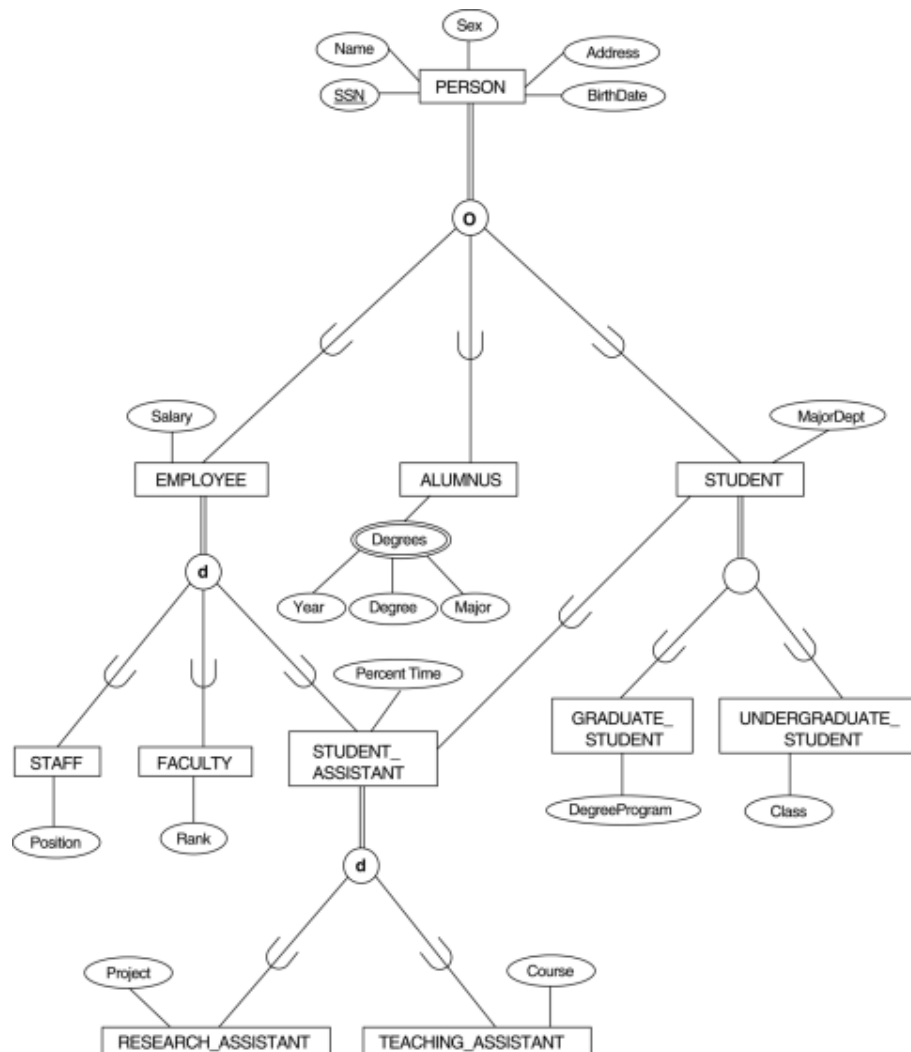


(d) PART

<u>PartNo</u>	Description	MFlag	DrawingNo	ManufactureDate	BatchNo	PFlag	SupplierName	ListPrice
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- Mapping of Shared Subclasses (Multiple Inheritance)
 - A shared subclass, such as STUDENT_ASSISTANT, is a subclass of several classes, indicating multiple inheritance. These classes must all have the same key attribute; otherwise, the shared subclass would be modeled as a category.
 - We can apply any of the options discussed in Step 8 to a shared subclass, subject to the restriction discussed in Step 8 of the mapping algorithm. Below both 8C and 8D are used for the shared class STUDENT_ASSISTANT.

Mapping EER Model Constructs to Relations (8)



PERSON

<u>SSN</u>	Name	BirthDate	Sex	Address
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EMPLOYEE

<u>SSN</u>	Salary	EmployeeType	Position	Rank	PercentTime	RAFlag	TAFlag	Project
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ALUMNUS

<u>SSN</u>	Year	Degree
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ALUMNUS_DEGREES

<u>SSN</u>	Year	Degree
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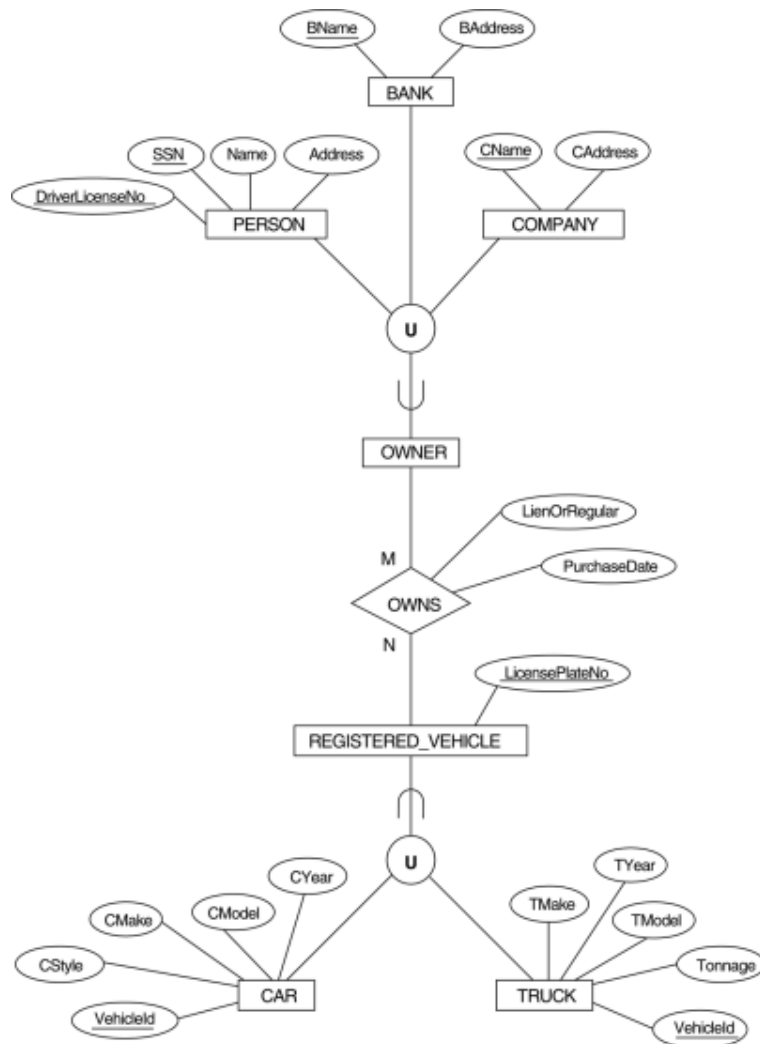
STUDENT

<u>SSN</u>	MajorDept	GradFlag	UndergradFlag	DegreeProgram	Class	StudAssistFlag
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- **Step 9: Mapping of Union Types (Categories).**

- For mapping a category whose defining superclass have different keys, it is customary to specify a new key attribute, called a surrogate key, when creating a relation to correspond to the category.
- In the example below we can create a relation OWNER to correspond to the OWNER category and include any attributes of the category in this relation. The primary key of the OWNER relation is the surrogate key, which we called OwnerId.

Mapping EER Model Constructs to Relations (9)



PERSON

<u>SSN</u>	DriverLicenseNo	Name	Address	
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BANK

<u>BName</u>	BAddress	OwnerId
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COMPANY

<u>CName</u>	CAddress	OwnerId
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OWNER

<u>OwnerId</u>

REGISTERED_VEHICLE

<u>VehicleId</u>	LicensePlateNumber
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CAR

<u>VehicleId</u>	CStyle	CMake	CModel	
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TRUCK

<u>VehicleId</u>	TMake	TModel	Tonnage	TYear
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OWNS

<u>OwnerId</u>	<u>VehicleId</u>	PurchaseDate	LienOrRegular
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Q & A

